Meme Coin Analysis Dashboard

Data Engineering Project Proposal



much data.
such pipeline.

IDSC501

Team members: TBD

Doge illustration: Icons8

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01 **Ideation** — Why meme coins?

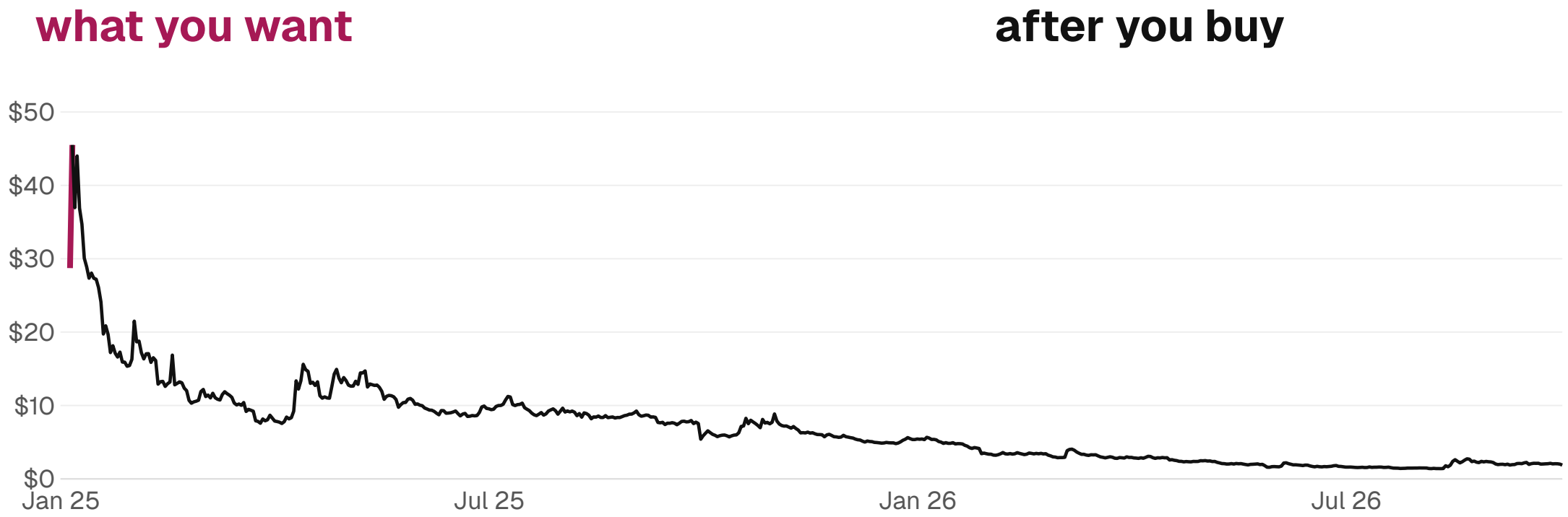
02 **Data collection** — Sources and access methods

03 **Preprocessing** — Records ready for analysis

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Why meme coins?

TRUMP daily close (USD)



What was the community saying?

18 Jan 2025–7 Oct 2026. Supplied daily closes. Captions are a joke, not an actual trade.

Data overview

Market data

Structured records

14,773

daily records across 10 coins

OHLCV and market cap
UTC date

Social data

Text + platform metadata

Posts + comments

Text and creation time
Observed reactions
Account or community

Prices available

Social collection planned

Price input ends on 7 Oct 2026. Selected X accounts and eligible Reddit communities define the social scope.

Data collection

	Source	Collection method	Output
	CoinMarketCap Historical Data	**Web crawling**	Daily price CSV
	X via FxTwitter	**Third-party API**	Posts + reactions in JSON
	Reddit via Arctic Shift	**Archive API / files**	Posts + comments

Daily social batches, with a seven-day overlap.

Project websites: crawl links and contract details to verify sources.

Reddit awaits access conditions. Price refresh needs new CSVs or restoration of the team crawler.

Data preprocessing

From raw posts to daily features.

Raw JSON

```
{
  "id": "<post_id>",
  "text": "$DOGE…",
  "created_at": "<UTC>",
  "likes": null
}
```

Standardize

Stable post ID
UTC timestamp
Language + repost flag
Verified coin match

Daily features

Coin + UTC date
Observed post count
Mean sentiment
Coverage status

Price checks Duplicate days, OHLC rules, missing or zero values

NULL ≠ 0 Collection failure stays missing.

Schema illustration. Sentiment uses supported English text. Preserve raw inputs for replay.

Database management

MongoDB

Social documents

social_items
Text + common fields

observations
Reactions + fetch time

Python ETL

Coin matching
Daily aggregation

PostgreSQL

Relational tables

price_daily
Prices by asset + date

social_daily
By asset + date + scope

Raw archive Original responses, source URLs, timestamps and checksums

PostgreSQL also supports JSON.

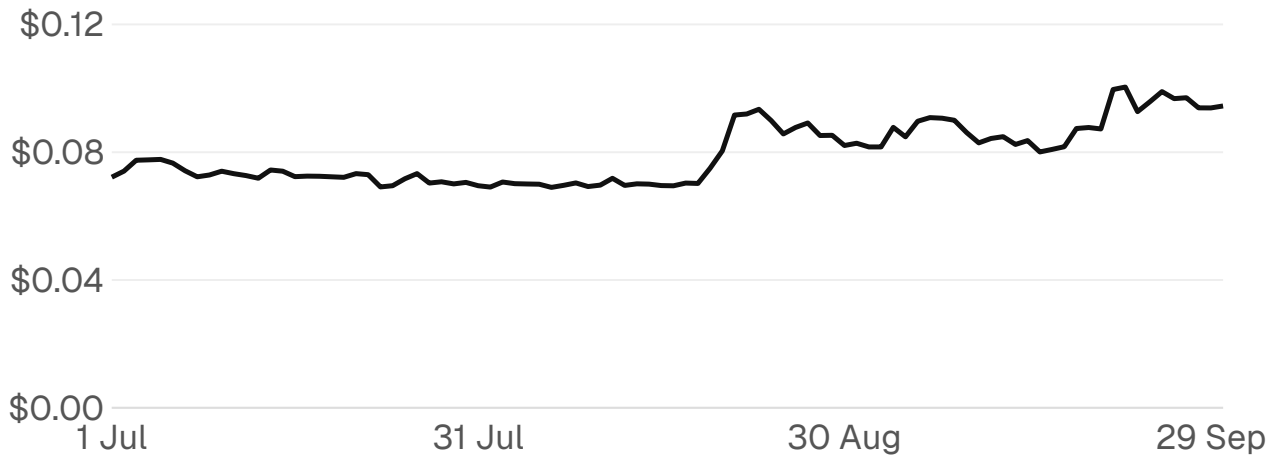
MongoDB separates social documents and makes the ETL explicit.

Price CSVs load into PostgreSQL after validation. Two stores add backup and consistency work.

Dashboard & expected outcome

Collect → Prepare → Store → Explore

DOGE 2026 Q3 Concept preview



Social activity

Awaiting eligible collection

Coverage: unavailable

Compare time series

Inspect posts

Check coverage

Streamlit + Plotly. Charts query PostgreSQL. Post details query MongoDB. DOGE prices use supplied data.